LINUX BOOTCAMP

Complete Command Reference
Days 1-5

Comprehensive Guide for Beginners

June 5, 2026

Contents

1 DAY 1: Filesystem & Navigation

1.1 Foundational Concepts

Key Concepts

- **Linux:** Operating system kernel and ecosystem
- **Filesystem:** Hierarchical tree structure starting from root (/)
- **Root Directory (/):** Top-level directory containing everything
- **Home Directory (~):** Personal user workspace
- **Absolute Path:** Starts with / (works from anywhere)
- **Relative Path:** No leading / (depends on current location)

1.2 Basic Navigation Commands

1.2.1 Print Working Directory

```
pwd
```

Purpose: Shows your current location in the filesystem

Example Output:

```
/home/ahmerijaz
```

1.2.2 List Directory Contents

```
ls
ls -l
ls -la
ls -lR
ls -lRa
```

Flags Explained:

- `-l` = Long format with detailed information
- `-a` = Show all files including hidden files (starting with .)
- `-R` = Recursive (show subdirectories)

1.2.3 Change Directory

```
cd /path/to/directory
cd ~
cd /
cd ..
cd ../..
cd -
```

Shortcuts Explained:

- `cd ~` = Go to home directory
- `cd /` = Go to root directory
- `cd ..` = Go up one level
- `cd ../..` = Go up two levels
- `cd -` = Go to previous directory

1.3 Linux Filesystem Structure

```
/                       # Root - top of filesystem
/home/                  # User home directories
/home/username/         # Individual user's files
/etc/                   # System configuration files
/var/                   # Variable data (logs, caches)
/var/log/               # System and application logs
/bin/                   # Essential commands/programs
/usr/                   # User programs and libraries
/tmp/                   # Temporary files
/proc/                  # Running process information
/root/                  # Root user's home directory
/dev/                   # Device files
/sys/                   # System information
```

1.4 Absolute vs Relative Paths

1.4.1 Absolute Paths (Start with /)

```
cd /home/ahmerijaz/bootcamp
cd /var/log
cd /etc/config
pwd
```

Characteristics:

- Start with forward slash (/)
- Work from anywhere on the system
- Complete path from root

1.4.2 Relative Paths (No leading /)

```
cd bootcamp
cd projects/cybersecurity
cd ..
cd ../..
cd ./folder
```

Characteristics:

- Relative to your current location

- Change meaning based on where you are
- Shorter to type

2 DAY 2: File Management

2.1 Creating Files and Directories

2.1.1 Create Empty File

```
touch filename.txt
touch file1.txt file2.txt file3.txt
```

Usage: Creates empty file or updates timestamp if exists

2.1.2 Create Directory

```
mkdir directoryname
mkdir folder1 folder2 folder3
```

2.1.3 Create Nested Directories

```
mkdir -p folder1/folder2/folder3/folder4
```

Flag -p: Creates parent directories automatically

2.2 Writing to Files

2.2.1 Overwrite File (DANGER!)

```
echo "content here" > filename.txt
```

Warning: The > operator **OVERWRITES** existing content!

2.2.2 Append to File (SAFE)

```
echo "more content" >> filename.txt
echo "another line" >> filename.txt
```

Note: The >> operator adds to end without deleting

2.2.3 Multi-line Writing

```
cat << EOF > filename.txt
Line 1 of content
Line 2 of content
Line 3 of content
EOF
```

2.3 Reading Files

2.3.1 Display Entire File

```
cat filename.txt
```

Usage: Displays complete file content to terminal

2.3.2 View Large Files Page by Page

```
less filename.txt
```

Navigation Keys:

- Space = Next page
- b = Previous page
- g = Beginning of file
- G = End of file
- / = Search
- q = Quit

2.3.3 View First/Last Lines

```
head filename.txt
head -n 20 filename.txt
tail filename.txt
tail -n 20 filename.txt
tail -f filename.txt
```

Flags:

- `-n 20` = Show 20 lines
- `-f` = Follow file (watch for updates)

2.4 Copying Files and Directories

2.4.1 Copy Single File

```
cp source.txt destination.txt
cp /home/user/file.txt /tmp/file.txt
```

2.4.2 Copy Multiple Files

```
cp file1.txt file2.txt file3.txt /destination/
cp *.txt /backup/
```

2.4.3 Copy Directory (Recursive)

```
cp -r source_folder destination_folder
cp -rv source_folder destination_folder
```

Flags:

- `-r` = Recursive (copies directory and contents)
- `-v` = Verbose (shows what's being copied)

2.5 Moving and Renaming Files

2.5.1 Rename File

```
mv oldname.txt newname.txt
mv file1.txt file2.txt file3.txt /destination/
```

2.5.2 Move File to Different Directory

```
mv file.txt /path/to/destination/
mv file.txt /path/to/destination/newname.txt
```

2.5.3 Move Multiple Files

```
mv *.txt /archive/
mv file1 file2 file3 /destination/
```

2.6 Deleting Files and Directories

2.6.1 Delete File

```
rm filename.txt
rm file1.txt file2.txt file3.txt
rm *.tmp
```

WARNING: No recycle bin! Deleted files are GONE forever!

2.6.2 Delete Directory

```
rmdir empty_folder
rm -r folder_with_contents
rm -rf folder_with_contents
```

Flags:

- `rmdir` = Only removes empty directories
- `-r` = Recursive (deletes directory and contents)
- `-f` = Force (don't ask for confirmation)

DANGER: `rm -rf /` would destroy your entire system!

3 DAY 3: File Permissions

3.1 Understanding Permissions

3.1.1 Permission Structure

```
-rw-r--r-- 1 owner group 1024 Jun 5 15:30 filename.txt
```

Breaking it down:

- `-` = File type (- = file, d = directory, l = link)
- `rw-` = Owner permissions (read, write, no execute)
- `r--` = Group permissions (read only)
- `r--` = Others permissions (read only)

3.1.2 Permission Meanings

Symbol	Name	Number	Meaning
r	Read	4	Can view/read file
w	Write	2	Can modify/edit file
x	Execute	1	Can run file as program
-	None	0	No permission

3.2 Viewing Permissions

3.2.1 View File Permissions

```
ls -l filename.txt
ls -la directory/
ls -l *.txt
```

Output Example:

```
-rw-r--r-- 1 ahmerijaz ahmerijaz 256 Jun 5 15:30 file.txt
```

3.2.2 View File Attributes

```
lsattr filename.txt
lsattr -R directory/
```

3.3 Changing Permissions with chmod

3.3.1 Using Symbolic Notation

```
chmod u+x file.txt
chmod g+w file.txt
chmod o-r file.txt
chmod a+r file.txt
chmod u=rwx,g=rx,o= file.txt
chmod -R 755 folder/
```

Symbols:

- `u` = User (owner)
- `g` = Group
- `o` = Others
- `a` = All
- `+` = Add permission
- `-` = Remove permission
- `=` = Set exactly

3.3.2 Using Numeric Notation

```
chmod 644 file.txt
chmod 755 script.sh
chmod 700 private_file.txt
chmod 600 secret.txt
chmod 640 group_readable.txt
chmod 444 read_only.txt
chmod 750 admin_script.sh
chmod -R 755 folder/
```

Number Breakdown:

- First digit = Owner (read + write + execute = 4+2+1 = 7)
- Second digit = Group
- Third digit = Others

3.3.3 Common Permission Values

Value	Symbolic	Usage
644	-rw-r–r–	Regular file (owner write, others read)
755	-rwxr-xr-x	Executable (owner full, others read/exec)
700	-rwx——	Private file (owner only)
600	-rw——-	Secret file (owner only)
640	-rw-r——	Restricted (owner write, group read)
444	-r–r–r–	Read-only (nobody can write)
750	-rwxr-x—	Admin script (owner+group execute)
777	-rwxrwxrwx	Everything allowed (DANGEROUS!)

3.4 File Attributes with chattr

3.4.1 Make File Immutable

```
sudo chattr +i filename.txt
```

Effect: File cannot be deleted, modified, renamed, or moved. Even root cannot touch it!

3.4.2 Make File Append-Only

```
sudo chattr +a filename.txt
```

Effect: Can only append data. Cannot modify or delete old content. Perfect for logs.

3.4.3 Remove Attributes

```
sudo chattr -i filename.txt
sudo chattr -a filename.txt
sudo chattr -ia filename.txt
```

3.4.4 View File Attributes

```
lsattr filename.txt
lsattr -la directory/
```

3.4.5 Common Attributes

Flag	Meaning
+i	Immutable (cannot be deleted/modified)
-i	Remove immutable flag
+a	Append-only (can only add data)
-a	Remove append-only flag
+d	No dump (skip in backups)
-d	Remove no-dump flag

4 DAY 4: Users & Groups

4.1 Understanding Users and Groups

Key Concepts

- **UID (User ID):** Unique number for each user (0 = root)
- **GID (Group ID):** Unique number for each group
- **Owner:** Single person who owns a file
- **Group:** Collection of users with shared permissions
- **Others:** Everyone else on the system
- **Root:** Special user with UID 0 (can do anything)

4.2 User Information Commands

4.2.1 Show Current User

```
whoami
```

Output: Your username (e.g., `ahmerijaz`)

4.2.2 Show User ID and Groups

```
id
id username
id analyst
id developers
```

Example Output:

```
uid=1000(ahmerijaz) gid=1000(ahmerijaz)
  ↪ groups=1000(ahmerijaz),27(sudo),4(adm)
```

Interpretation:

- `uid=1000` = You are user ID 1000
- `gid=1000` = Your primary group is ID 1000
- `groups=...` = All groups you belong to

4.2.3 Show Groups User Belongs To

```
groups
groups username
groups analyst
```

4.3 User Management

4.3.1 Create New User

```
sudo useradd username
sudo useradd -m username
sudo useradd -m -s /bin/bash username
sudo useradd -m -s /bin/bash analyst
sudo useradd -m -s /bin/bash developer
```

Flags:

- `-m` = Create home directory
- `-s /bin/bash` = Set login shell to bash

4.3.2 Verify User Creation

```
id analyst
id developers
cat /etc/passwd | grep analyst
```

4.3.3 Delete User

```
sudo userdel username
sudo userdel -r username
```

Flags:

- (no flag) = Delete user only, keep home directory
- `-r` = Delete user AND home directory

4.4 Group Management

4.4.1 Create Group

```
sudo groupadd groupname
sudo groupadd security
sudo groupadd tools
sudo groupadd developers
```

4.4.2 Verify Group Creation

```
cat /etc/group
cat /etc/group | grep security
cat /etc/group | grep -E "security|tools"
```

4.4.3 Delete Group

```
sudo groupdel groupname
sudo groupdel security
```

4.4.4 Add User to Group

```
sudo usermod -aG groupname username
sudo usermod -aG security analyst
sudo usermod -aG tools developers
```

Flags:

- `-a` = Append (add to group, don't replace)
- `-G` = Groups

4.4.5 Verify User in Group

```
id analyst
id developers
groups analyst
```

4.5 File Ownership

4.5.1 Change File Owner

```
sudo chown newowner filename.txt
sudo chown analyst security_report.txt
sudo chown developers development_log.txt
```

4.5.2 Change File Group

```
sudo chown :newgroup filename.txt
sudo chown :security filename.txt
sudo chown :tools development_log.txt
```

Note: The colon (:) means "change group only, keep owner same"

4.5.3 Change Both Owner and Group

```
sudo chown owner:group filename.txt
sudo chown analyst:security security_report.txt
sudo chown developers:tools development_log.txt
sudo chown ahmerijaz:ahmerijaz public_notes.txt
```

4.5.4 Change Ownership Recursively

```
sudo chown -R owner:group directory/
sudo chown -R analyst:security /home/ahmerijaz/bootcamp/
```

Flag -R: Applies changes to all files and subdirectories

4.5.5 Verify Ownership Changed

```
ls -l filename.txt
ls -l *.txt
ls -la directory/
```

4.6 Using sudo

4.6.1 Run Single Command as Root

```
sudo command
sudo apt update
sudo chmod 600 filename.txt
sudo chown user:group filename.txt
```

Process:

1. System asks: `[sudo] password for ahmerijaz:`
2. Enter YOUR password (not root's!)
3. Command runs with root privileges

4.6.2 Become Root User

```
sudo su
```

Effect: You are now the root user (UID 0)
WARNING: Be extremely careful! You can destroy the system!

4.6.3 Exit Root Shell

```
exit
```

4.6.4 Check if User Can Use sudo

```
groups
```

If you see `sudo` in the output, you can use sudo

4.7 User and Group Files

4.7.1 View User Database

```
cat /etc/passwd
cat /etc/passwd | grep username
```

Format: `username:x:UID:GID:Full Name:Home:/bin/bash`

4.7.2 View Password Hashes (Root Only)

```
sudo cat /etc/shadow
```

4.7.3 View Group Database

```
cat /etc/group
cat /etc/group | grep groupname
```

Format: `groupname:x:GID:members`

5 DAY 5: Finding Files with find

5.1 Introduction to find

find Overview

The `find` command searches the filesystem for files and directories based on various criteria. It's recursive (searches subdirectories) and can execute commands on results.

Basic Syntax:

```
find /path [criteria] [action]
```

5.2 Find by Name

5.2.1 Find Exact Filename

```
find /path -name "filename.txt"
find /home/ahmerijaz/bootcamp -name "security_report.txt"
```

5.2.2 Find with Wildcards

```
find /path -name "*.txt"
find /path -name "file*.txt"
find /path -name "*report*"
find /home/ahmerijaz/bootcamp -name "*.txt"
```

Wildcard Meanings:

- `*` = Match anything (any characters)
- `?` = Match single character
- `[abc]` = Match a, b, or c

5.2.3 Case-Insensitive Search

```
find /path -iname "*.TXT"
find /path -iname "FILE*"
find /home/ahmerijaz/bootcamp -iname "*.TXT"
```

Note: The `-i` in `-iname` makes search case-insensitive

5.3 Find by Type

5.3.1 Find Files Only

```
find /path -type f
find /home/ahmerijaz/bootcamp -type f
```

5.3.2 Find Directories Only

```
find /path -type d
find /home/ahmerijaz/bootcamp -type d
```

5.3.3 Find Symbolic Links

```
find /path -type l
find / -type l
```

5.3.4 Find Sockets and Pipes

```
find /path -type s
find /path -type p
```

Type Reference:

Type	Meaning
f	Regular file
d	Directory
l	Symbolic link
s	Socket
p	Named pipe
b	Block device
c	Character device

5.4 Find by Size

5.4.1 Find Large Files

```
find /path -type f -size +100c
find /path -type f -size +1M
find /path -type f -size +100M
find /home/ahmerijaz/bootcamp -type f -size +1M
```

5.4.2 Find Small Files

```
find /path -type f -size -100c
find /path -type f -size -1M
find /home/ahmerijaz/bootcamp -type f -size -100c
```

5.4.3 Find Exact Size

```
find /path -type f -size 100c
find /path -type f -size 1M
```

Size Units and Operators:

Symbol	Meaning	Example
+	Larger than	-size +100c
-	Smaller than	-size -100c
(none)	Exactly	-size 100c
c	Bytes	-size 100c
k	Kilobytes	-size 1k
M	Megabytes	-size 1M
G	Gigabytes	-size 1G

5.5 Find by Permissions

5.5.1 Find World-Writable Files

```
find /path -type f -perm -002
find / -type f -perm -002
find /home -type f -perm -002
```

Security Check: World-writable files are potential vulnerabilities!

5.5.2 Find Readable by Group

```
find /path -type f -perm -040
find /home/ahmerijaz/bootcamp -type f -perm -040
```

5.5.3 Find Executable Files

```
find /path -type f -executable
find /home/ahmerijaz/bootcamp -type f -executable
find / -type f -executable
```

5.5.4 Find SUID Files

```
find / -type f -perm -4000
```

Security: SUID files run as owner (potential privilege escalation)

5.6 Find by Owner

5.6.1 Find Files Owned by User

```
find /path -type f -user username
find /home/ahmerijaz/bootcamp -type f -user analyst
find /home/ahmerijaz/bootcamp -type f -user developers
find / -type f -user root
```

5.6.2 Find Files Owned by Group

```
find /path -type f -group groupname
find /home/ahmerijaz/bootcamp -type f -group security
find /home/ahmerijaz/bootcamp -type f -group tools
```

5.6.3 Find Files with No Owner

```
find / -type f -nouser
find / -type f -nogroup
```

5.7 Find by Modification Time

5.7.1 Find Recently Modified Files

```
find /path -type f -mmin -60
find /path -type f -mmin -30
find /home/ahmerijaz/bootcamp -type f -mmin -60
```

Example: `-mmin -60` = Modified within last 60 minutes

5.7.2 Find Old Files

```
find /path -type f -mmin +60
find /path -type f -mtime +7
find / -type f -mtime +30
```

Time Units:

- `-mmin` = Modification minutes
- `-mtime` = Modification days
- `-atime` = Access time days
- `-ctime` = Change time days

5.7.3 Time Operators

Operator	Meaning
+N	More than N units ago
-N	Less than N units ago (within)
N	Exactly N units ago

5.8 Find by Empty

5.8.1 Find Empty Files and Directories

```
find /path -type f -empty
find /path -type d -empty
find /home/ahmerijaz/bootcamp -type f -empty
```

5.9 Combining Multiple Conditions

5.9.1 AND Logic (All conditions must be true)

```
find /path -type f -name "*.txt" -user ahmerijaz
find /path -type f -size +1M -mmin -60
find /path -type f -name "*.log" -size -100c
find /home/ahmerijaz/bootcamp -type f -name "*.txt" -size -100c
find /home/ahmerijaz/bootcamp -type f -name "*.txt" -user ahmerijaz
```

Note: Multiple conditions are joined with AND by default

5.10 Find Actions

5.10.1 Display Results with Details

```
find /path -type f -ls
find /path -name "*.txt" -ls
find /home/ahmerijaz/bootcamp -type f -ls
```

Output: Same format as `ls -l` with file details

5.10.2 Execute Command on Results

```
find /path -name "*.txt" -exec cat {} \;
find /path -name "*.tmp" -exec rm {} \;
find /path -type f -exec chmod 644 {} \;
find /home/ahmerijaz/bootcamp -name "security_report.txt" -exec ls
  ↪ -l {} \;
```

Syntax:

- `-exec` = Execute command
- {} = Placeholder for filename
- \; = End of command (escaped semicolon)

5.10.3 Delete Files (DANGEROUS!)

```
find /path -name "*.tmp" -delete
find /path -type f -empty -delete
```

WARNING: `-delete` removes files permanently! Use with extreme caution!

5.10.4 Prompt Before Action

```
find /path -name "*.txt" -ok rm {} \;
```

Difference: `-ok` asks for confirmation, `-exec` doesn't

5.11 Limit Search Depth

5.11.1 Search Only N Levels Deep

```
find /path -maxdepth 1 -type f
find /path -maxdepth 2 -type f
find /home/ahmerijaz/bootcamp -maxdepth 2 -type f
```

Example:

- `-maxdepth 1` = Current directory only
- `-maxdepth 2` = Current + 1 subdirectory level

5.11.2 Search Starting from N Levels

```
find /path -mindepth 2 -type f
```

5.12 Real-World Security Examples

5.12.1 Find Recently Modified Files (Breach Investigation)

```
find / -type f -mmin -60
find / -type f -mmin -30
find / -type f -mtime -1
```

5.12.2 Find World-Writable Files (Vulnerability Audit)

```
find / -type f -perm -002
find /home -type f -perm -002
find /var/www -type f -perm -002
```

5.12.3 Find SUID Files (Privilege Escalation Check)

```
find / -type f -perm -4000
```

5.12.4 Find Executable Scripts

```
find /home -type f -executable
find / -type f -name "*.sh" -executable
```

5.12.5 Find Large Files (Disk Space Analysis)

```
find / -type f -size +100M
find / -type f -size +1G
```

5.12.6 Find Suspicious Files

```
find / -type f -name "*.backdoor"
find / -type f -name "*shell*"
find / -type f -name "*nc*" -executable
```

5.12.7 Find Files Modified by Specific User

```
find / -type f -user suspicious_user -mtime -1
```

6 Quick Reference Tables

6.1 Common chmod Values

Value	Symbolic	Numeric	Use Case
644	-rw-r–r–	6-4-4	Regular file (default)
755	-rwxr-xr-x	7-5-5	Executable script
700	-rwx——	7-0-0	Private (owner only)
600	-rw——-	6-0-0	Secret file (password)
640	-rw-r—–	6-4-0	Group readable
444	-r–r–r–	4-4-4	Read-only
750	-rwxr-x—	7-5-0	Admin script
777	-rwxrwxrwx	7-7-7	DANGEROUS!

6.2 Permission Number Reference

Number	Symbolic	Permissions
0	—	No permissions
1	–x	Execute only
2	-w-	Write only
3	-wx	Write + Execute
4	r–	Read only
5	r-x	Read + Execute
6	rw-	Read + Write
7	rwx	Read + Write + Execute

6.3 find Command Criteria

Criteria	Syntax	Example
Name	-name "pattern"	-name "*.txt"
Type	-type [f/d/l]	-type f
Size	-size [+/-]N[c/k/M/G]	-size +100M
Owner	-user username	-user analyst
Group	-group groupname	-group security
Permissions	-perm mode	-perm -002
Modification	-mmin/-mtime N	-mmin -60
Empty	-empty	-empty
Executable	-executable	-executable

6.4 Useful Linux Shortcuts

Shortcut	Meaning
˜	Home directory
.	Current directory
..	Parent directory
-	Previous directory
/	Root directory
*	Match any characters
?	Match single character
˜username	Another user's home

7 Important Files and Directories

File/Directory	Purpose
/etc/passwd	User account information
/etc/shadow	Password hashes (root only)
/etc/group	Group information
/etc/sudoers	Sudo access configuration
/var/log	System and app logs
/var/log/auth.log	Authentication events
/var/log/syslog	System messages
/home	User home directories
/root	Root user's home
/tmp	Temporary files
/bin	Essential user commands
/usr/bin	User programs
/usr/local/bin	Locally installed programs
/dev	Device files
/proc	Process information

8 Complete Command Summary

8.1 Day 1: Navigation

```
pwd                      # Current directory
cd /path                 # Change directory
cd ~                     # Home directory
cd /                     # Root directory
cd ..                    # Parent directory
cd ../..                 # Two levels up
ls                       # List files
ls -l                    # Long format
ls -la                   # Long + hidden files
```

8.2 Day 2: File Operations

```
touch file.txt           # Create empty file
mkdir folder             # Create directory
mkdir -p a/b/c           # Create nested folders
cp source dest           # Copy file
cp -r src dest           # Copy directory
mv old new               # Move/rename
rm file                  # Delete file
rm -r folder             # Delete directory
cat file                 # Display file
echo "text" > file       # Write (overwrite)
echo "text" >> file      # Write (append)
less file                # Page through file
```

8.3 Day 3: Permissions

```
ls -l                    # View permissions
chmod 644 file           # Change permissions
chmod u+x file           # Add execute to owner
chmod -R 755 folder      # Recursive chmod
chown user:group file    # Change ownership
chown -R user:group folder # Recursive chown
chattr +i file           # Make immutable
chattr -i file           # Remove immutable
chattr +a file           # Append-only
lsattr file              # View attributes
```

8.4 Day 4: Users and Groups

```
whoami                   # Current user
id                       # UID and groups
id username              # User info
groups                   # User's groups
sudo useradd -m -s /bin/bash user    # Create user
```

```
sudo groupadd group     # Create group
sudo usermod -aG group user          # Add to group
sudo chown user:group file           # Change owner
sudo chown -R user:group folder      # Recursive
sudo su                 # Become root
cat /etc/passwd         # View users
cat /etc/group          # View groups
```

8.5 Day 5: Finding Files

```
find /path -name "file.txt"          # Find by name
find /path -name "*.txt"             # Find by pattern
find /path -type f                   # Find files
find /path -type d                   # Find folders
find /path -size +100M               # Find large
find /path -user username            # Find by owner
find /path -perm -002                # World-writable
find /path -mmin -60                 # Recently modified
find /path -type f -empty            # Find empty
find /path -type f -executable       # Find executable
find /path -name "*.txt" -exec cat {} \;  # Execute
find /path -maxdepth 2 -type f       # Limit depth
```

9 Security Checklist

9.1 File Security Best Practices

- ✓ Passwords/private keys should be 600 (-rw——-)
- ✓ Regular files should be 644 (-rw-r–r–)
- ✓ Executable scripts should be 755 (-rwxr-xr-x)
- ✓ No world-writable files in system directories
- ✓ Critical files should have immutable flag (+i)
- ✓ Logs should have append-only flag (+a)
- ✓ System config files should be owned by root
- ✓ User files should be owned by respective user

9.2 User and Group Security

- ✓ Limit who can use sudo (use sudoers file)
- ✓ Use sudo for specific commands, not full access
- ✓ Remove unused user accounts
- ✓ Keep sudo group small
- ✓ Review /var/log/auth.log regularly
- ✓ Disable unnecessary user accounts
- ✓ Use strong passwords
- ✓ Check for orphaned files (files with no owner)

9.3 Audit Commands

```
# Security Audits
find / -type f -perm -002                  # World-writable
find / -type f -perm -4000                 # SUID files
find / -type f -mmin -60                   # Recently modified
find / -type f -empty                      # Empty files
find / -type f -nouser                     # No owner
find / -type f -nogroup                    # No group
cat /etc/passwd                            # All users
cat /etc/group                             # All groups
sudo cat /var/log/auth.log | tail -20   # Recent logins
```

Notes

This document covers the fundamental Linux commands taught in Days 1-5 of the Linux Bootcamp. All commands have been tested and verified to work correctly on Ubuntu 24.04 systems.

Key Takeaways:

1. Understanding the filesystem is essential
2. File permissions control who can access what
3. Users and groups organize access control
4. find searches for files by any criteria
5. Security requires proper permissions and monitoring